Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP090B002-V1

Revision Record

Rev No.	Date	Description
A/0	2023-05-09	First Release

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Revision Record

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1.General Specifications

No.	Item	Specification	Unit	Remark
1	LCD Size	0.9”	inch	-
2	Panel Type	IPS	-	-
3	Resolution	128x RGB x 128	Pixel	-
4	Display Mode	Normally Black		-
5	Number of Colors	65K	-	-
6	Viewing Direction	All	-	Note1
7	NTSC		-	Typ.
8	Contrast Ratio		-	
9	Luminance	TBD	cd/m2	Typ.
10	Module Size	25.24(H)x27.14(V)x1.55(D)	mm	Note1
11	Panel Active Area	23.04(H)x23.03(V)	mm	Note1
12	Pixel Pitch		mm	-
13	Pixel Arrangement	RGB-stripe		-
14	Weight	TBD	g	-
15	Driver IC	ST7735	-	-
16	Driver IC RAM Size		bit	-
17	Light Source	2 White LEDs	-	-
18	Interface	SPI	-	-
19	Operating Temperature	-20~+70	℃	-
20	Storage Temperature	-30~+80	℃	-

Note 1: Please refer to the mechanical drawing；

2.Pin Assignments

No.	Symbol	Function
1	GND	Ground.
2	LEDK	Backlight cathode.
3	VDD	Power Supply for Logic. This is a voltage supply pin. It must be connected to external source.
4	DC	Data is regarded as a parameter or data when SCL is high
5	CS	Chip selection signal.
6	SCK	Serial input CLK
7	SI	Source Driver Output Pins.
8	RESET	Chip reset signal.
9	LEDA	Backlight anode.
10	NC	NC.

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3. Electrical Specification

3.1 Absolute Maximum Ratings

Item	Symbol	Rating	Unit
Supply Voltage	VDD	- 0.3 ~ +4.6	V
Supply Voltage (Logic)	VDDI	- 0.3 ~ +4.6	V
Supply Voltage (Digital)	VCC	-0.3 ~ +1.95	V
Driver Supply Voltage	VGH-VGL	-0.3 ~ +30.0	V
Logic Input Voltage Range	VIN	-0.3 ~ VDDI + 0.3	V
Logic Input Voltage Range	VO	-0.3 ~ VDDI + 0.3	V

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3.2 Typical Operation Conditions

Parameter	Symbol	Condition	Specification			Unit	Related Pins
			Min	Typ	Max		
Power & Operation Voltage							
System Voltage	VDD	Operating Voltage	2.5	2.75	3.7	V	
Interface Operation Voltage	VDDI	I/O Supply Voltage	1.65	1.8	3.7	V	
Gate Driver High Voltage	VGH		11		16	V	Note 4
Gate Driver Low Voltage	VGL		-13		-7.5	V	
Gate Driver Supply Voltage		| VGH-VGL |	18.5		29	V	Note 4
Input / Output							
Logic-High Input Voltage	VIH		0.7VDDI		VDDI	V	Note 1
Logic-Low Input Voltage	VIL		VSS		0.3VDDI	V	Note 1
Logic-High Output Voltage	VOH	IOH = -1.0mA	0.8VDDI		VDDI	V	Note 1
Logic-Low Output Voltage	VOL	IOL = +1.0mA	VSS		0.2VDDI	V	Note 1
Logic-High Input Current	IIH	VIN = VDDI			1	uA	Note 1
Input eakage							
VCOM Voltage							
VCOM Amplitude	VCOM		-2		-0.425	V	
Source driver							
Source Output Range	Vsout		0.1		GVDD	V	
Gamma Reference Voltage	GVDD		3.15		4.7	V	
Source Output Settling Time	Tr	Below with 99% precision			20	us	Note 2
Output Offset Voltage	Voffset				35	mV	Note 3

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3.3 Backlight Circuit Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current	I_B		40	-	mA
LED Voltage	Vf		3.2		V
Power Consumption	P_{BL}	-		-	mW

3.4 LCD Current Consumption

Item	Symbol	Typ.	Max.	Unit
Full Mode	VCI	-	-	mA
Test Condition：VCI=2.8V，IOVCC=2.8V； Interface Driver type ： Row flip or column flip； TN Type=>All Black Pattern. IPS Type=>All White Pattern. Temperature：25℃.				
Sleep Mode	VCI	-	-	uA
Test Condition：VCI=2.8V，IOVCC=2.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. Temperature：25℃.				

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4.Optical Specification

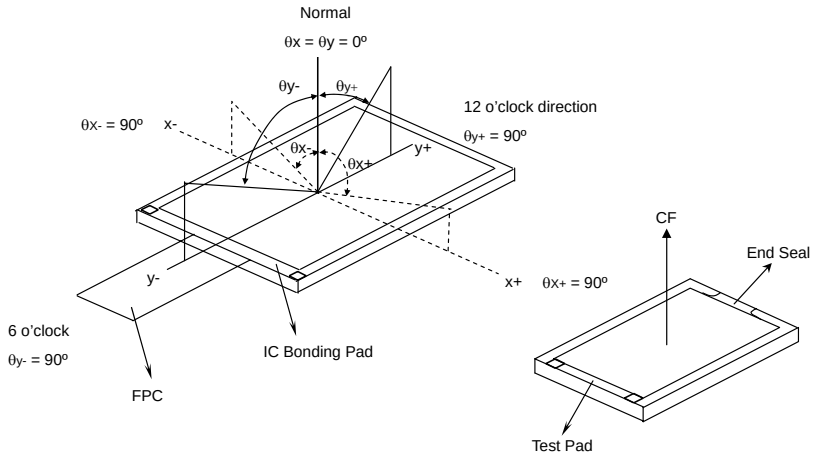
4.1 LCM Optical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	CR≧10	-		-	degree
	Right	θ_R		-		-	
	Top	θ_T		-		-	
	Bottom	θ_B		-		-	
Response Time 响应时间		Ton+Toff	θ=Φ=0°	-			ms
Contrast Ratio 对比度		CR	θ=Φ=0°			-	-
Luminance 亮度		L	θ=Φ=0°	-	250	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal θ=Φ=0°	-0.02		+0.02	-
		W_y					
	Red	R_x					
		R_y					
	Green	G_x					
		G_y					
	Blue	B_x					
		B_y					
Uniformity 均匀度		U_L	θ=Φ=0°	80	-	-	%
Flicker 闪烁		-	-	No Visible			-

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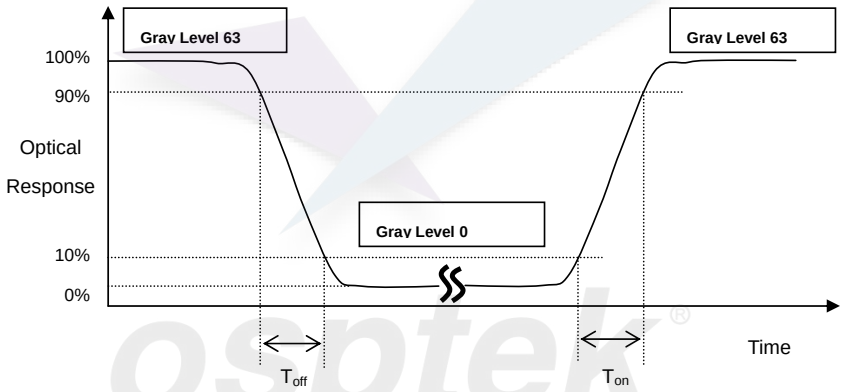
4.2 Measurement system

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

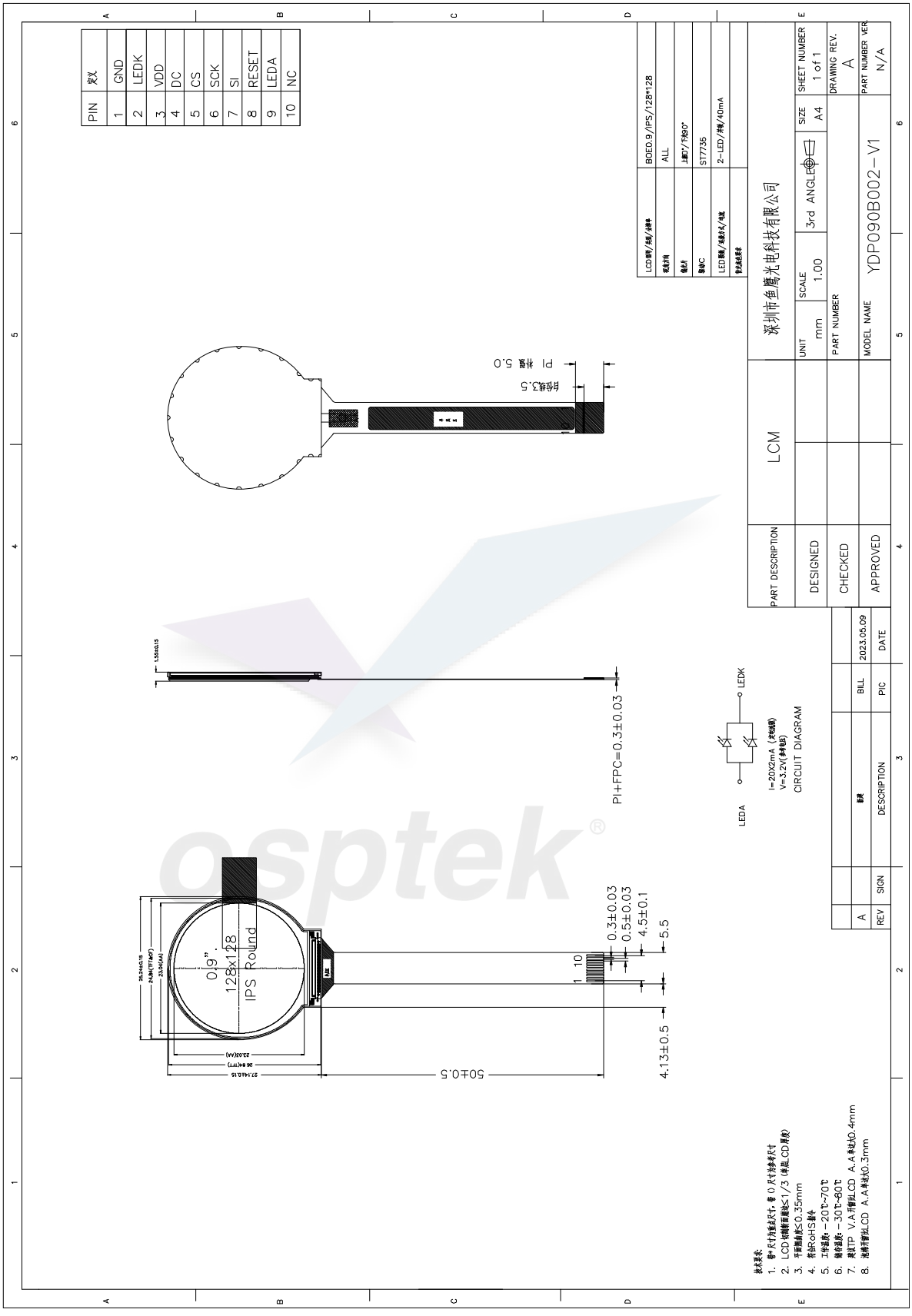
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing



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6. Reliability Test Items

Test Item	Test Condition	Test result determinant gist
High temperature storage	80±3℃，24H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the LCD; 2.Non-display; 3.Glass crack; 4.The electrical characteristics requirements shall be satisfied.
Low temperature storage	-30±3℃，24H；	
High temperature operation	70±3℃，24H；	
Low temperature operation	-20±3℃，24H；	
High temperature /humidity	60℃±3℃,90%±3%RH，24H；	
Thermal Shock	-30℃/0.5h~+80℃/0.5h for a total 24 cycles；	
Vibration Test	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H；(Packing condition)	
ESD test	±2KV, Human Body Mode, 150pF/330Ω； ±8KV, Air Mode, 150pF/330Ω；	

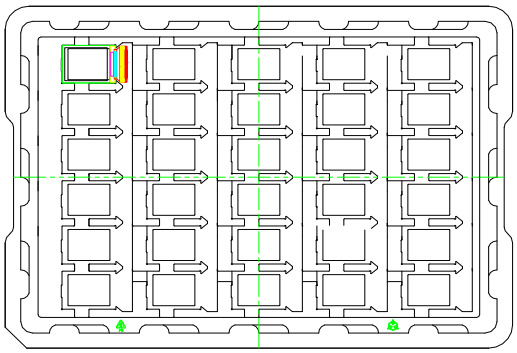
Remark:

1. The test samples should be applied to only one test item.
2. Sample size for each test item is 2pcs.
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

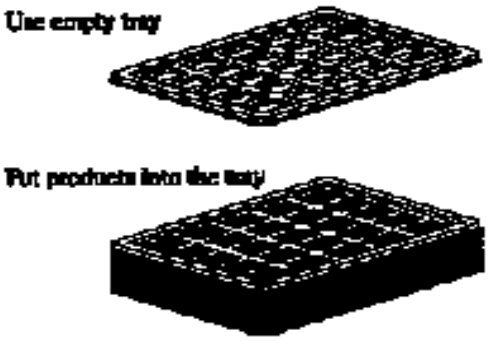
7. Packing and Storage Specification(Reference Only)

7.1 Packing Method

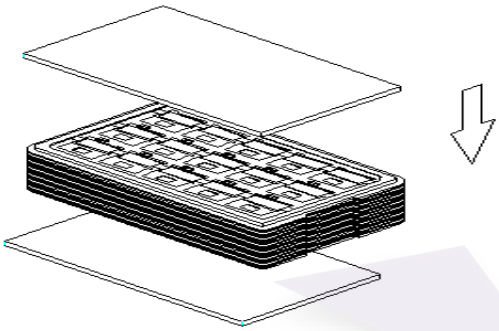
(1)



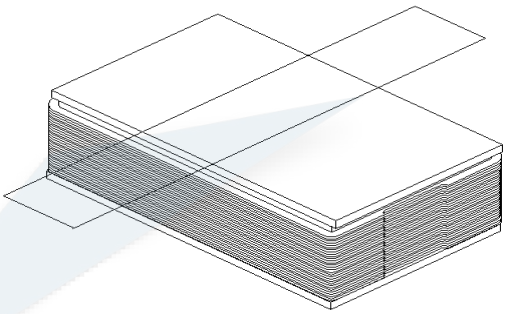
(2)



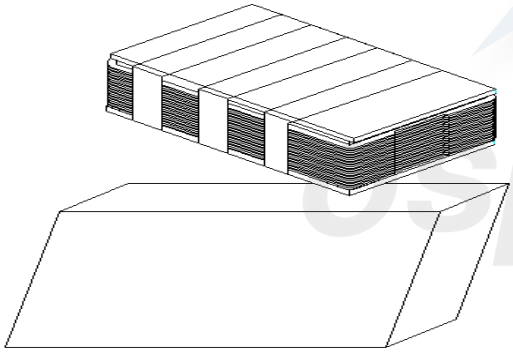
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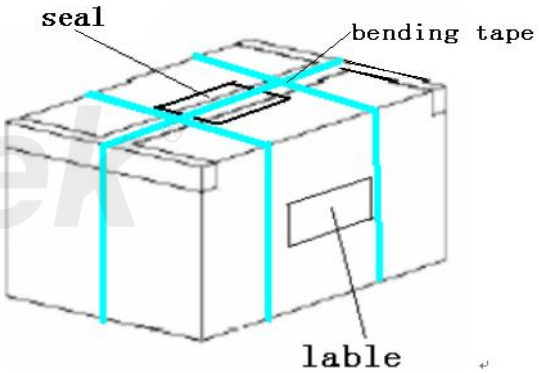
(4)



(5)



(6)



1. Put module into tray cavity. .
2. Tray stacking. .
3. Put 1 foam under the tray stack and 1 foam above. .
4. Fix the cardboard to the tray stack with adhesive tape. .
5. Put the tray stack into carton. .
6. Carton sealing with adhesive tape. .

7.2 Storage Method

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.

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8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.
5. Under the specified electrical parameters, the ambient temperature is 23 ° C ± 5 ° C, the humidity is 55%± 15%, and the hardware is not damaged. The normal service life is 30,000-50,000 hours.

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